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 TOPICAL REVIEW

Graph Databases for Group Decision Making in Industry: A Comprehensive Literature Review

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ABSTRACT Virtual manufacturing, simulation, and optimization provide a wealth of knowledge about the possibilities of future production systems to support decision makers. However, this knowledge usually remains with a handful of domain experts, is hard to share even within the same team and cannot be easily used to support further optimization activities. At the same time, simulations can benefit from the incorporation of linked data from real factories once a process is running, to improve the accuracy of the simulation and hence the accuracy of optimization results. Graph databases provide a possible approach to storing and managing this form of interrelated heterogeneous data, with powerful querying capabilities that can identify important or interesting patterns that might otherwise remain hidden. This paper provides a structured literature review of the current directions of research within this broad but interconnected set of topics, with a special focus on how graphs can support finding knowledge within Virtual Factories, used by larger teams for industrial planning and optimization. Papers have been selected from a wide number of research databases and reviewed from the perspectives of the industries they focus on, the technologies they focus on and the approach they have taken. We conclude that current research focuses on one or two aspects of this problem but does not address all at once, despite the potential benefits of the combination.

INDEX TERMS Graph database, optimization, simulation, knowledge graphs, industry 4.0.

I. INTRODUCTION

Since the advent of computer control, industrial production systems have increasingly used digital technologies and networking to manage their operations. The diverse mix of automation tools, machines, sensors, and processes that are involved in these production systems results in a wide variety of heterogeneous data, which is increasingly being collected and stored automatically. The advent of Industry 4.0 has led to a further increase in networking and the volume of data that can be accessed in a modern factory. This has led to further increases in the complexity and heterogeneity of that data, and we expect this increase to continue into the future [1].

Parallel to the use of computers and networking for managing data, computers have been used to assist in

planning and designing industrial products and systems, dating back to the earliest CAD systems. A range of technologies have emerged that extend the use of computer software for designing, analyzing, predicting, and optimizing engineering systems including: simulation, virtual commissioning, digital twins, and a broad range of visualization, analytics, and optimization tools. These tools are often referred to under the umbrella of Virtual Manufacturing, which can be “*defined as the creation of a virtual model*” [2]. These virtual models interact with data sources in two ways, both basing their parameters on data from the factory (most obviously seen in optimization models and Digital Twins) and potentially providing simulated example data that mimics factory data. At the same time, the creation of virtual models and the inferences and results of such models are expanding from being activities performed by a single individual into being iterative projects carried out by groups of people.

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The expansion of the size of teams working on virtual models introduces new coordination problems that are addressed by the field of Group Decision Making and the related tools for Group Decision Support (GDS). The methods available so far for group decision support based on Multi-Objective Optimization (MOO) have a few shortcomings. They do not consider the relationship between the variable space, consisting of the values of decision variables in the optimization, and the objective space, which comprises the objective functions to be optimized. In [3], the first Decision Support System (DSS) to bridge the variable and objective space for Multi-Objective Optimization Problems (MOOP) was built. However, it does not address groups of decision-makers. Also, many Group Decision Support Systems (GDSS) assume a limited set of solutions for decision-making [4], [5]. In contrast, the set of solutions created by a modern optimization method can be extensive, which requires additional exploration before an informed decision can be made [6]. Lastly, it seems each optimization problem and the decisions connected to it are treated as a single occurrence, with possible connections to other problem instances being neglected.

A. INTRODUCTION TO VF-KDO

The VF-KDO (Virtual Factories with Knowledge-Driven Optimization) [7] concept investigates the possibilities for enhancing optimization potential in factory design and operation, through examination of the results of virtual models and the identification of higher-level patterns and knowledge within those results. Extending simulation-based optimization projects to team-scale operations necessitates expanding the VF-KDO concept into the field of GDSS and developing techniques to capture the complex relationships between elements of derived knowledge. To this end, we have identified the use of knowledge graphs (KGs) and graph database applications as a promising approach for modeling the relationships between simulation models, optimization runs, and the derived knowledge. The use of KGs also provides the potential for identifying deeper relationships and deriving further knowledge from the virtual factory models.

Several researchers have begun to address the possible use of graph models for GDS. However, none of those that we have found appear to directly address themselves to the use of graph databases, preferring to focus on the use of graph models themselves, such as the earlier work of [8] and [9]. Other more recent work extends the use of the graph model with machine learning through the application of Graph Neural Networks to fuzzy models [10]. Where databases are used, and there is a connection to a graph, a separation is maintained between the graph model and the storage mechanism, for example in [11], where business decision processes are represented as graph processes but stored within MongoDB, thus not making the graphs themselves directly accessible.

The concept of virtual factories can provide the ability to assess performance at multiple levels and generate analytics data, enabling a decision maker to look at the overall factory performance and focus attention on analysis at the equipment/device level [12]. The term virtual factory (VF), sometimes used interchangeably with virtual copy, digital factory or digital twin, can therefore be defined as “an integrated simulation model of major sub-systems in a factory that considers the factory as a whole and provides an advanced decision support capability” [13]. Virtual factories, optimization models, knowledge, and decisions are connected through multiple intrinsic relationships. Knowledge is central to any decision-making process: the more a decision-maker understands the relationship between how to design/run a system and how it performs, the better the decisions that are made [14]. Therefore, the concept of “know why” – to acquire the level of knowledge about the processes at the scientific level with a complete model that is able to accurately predict the performance of a design/configuration over a wide range of conditions – represents the core knowledge needed to make optimal decisions [15]. While it seems trivial to relate virtual models with optimization and decision-making, simulation models are, in fact, evaluative by nature, i.e., instead of generating any optimal solutions by themselves, they can only provide performance evaluations of different design alternatives and/or courses of action. It is only when the virtual models are connected to optimization algorithms that optimal solutions can be sought automatically by Simulation-Based Optimization (SBO) technologies.

SBO methods are not the only approach to performing optimization tasks in industry. They are, as stated, structured with the model first, and then optimization experiments are carried out on the model. By contrast, agent-based AI tools offer a more tightly coupled approach, where simulation and optimization capabilities are deeply integrated. In such systems, each agent can make localized decisions, allowing for decentralized optimization, which is more suitable in environments with multiple autonomous actors than human decision support. On the other hand, self-optimizing tools such as hyper-heuristics represent another evolution in optimization strategy. While not the focus of this research, the VF-KDO approach would be readily able to incorporate AI models from self-optimizing systems such as hyper-heuristics or Deep Reinforcement Learning (DRL) agents trained via simulation [16]. These agents generate valuable policy knowledge and decision patterns through iterative learning, which could also be captured in a knowledge graph to enhance traceability, reuse, and human interpretability to support high-level human decision-making further.

SBO methodologies are capable of effectively searching the vast number of possible solutions and identifying the ones that optimize a given objective. Where there exists more than one objective, MOO algorithms are used to generate optimal trade-off solutions for the decision maker to choose from [17].

Since most real-world problems, such as production systems development, are inherently multi-criteria by nature, they inevitably involve multiple optimization objectives [18]. These MOO algorithms have been increasingly used in practice during the last two decades. A relatively recent research area connected with MOO is the creation and utilization of domain-specific knowledge during runtime. The trade-off optimal solutions generated through MOO may contain hidden clues about what constitutes a good solution. Data Mining (DM) methods can help reveal patterns that help the decision-maker gain a better understanding of the problem. The knowledge gained through DM can also be applied to future optimization scenarios. This process of generating and utilizing knowledge within MOO is an active research area commonly referred to as the KDO aspect of the VF-KDO concept.

The term KDO was first introduced in [19]. It refers to a framework for decision-makers to gain knowledge generated by multi/many-objective optimization algorithms, which can acquire and incorporate the knowledge and preferences of the decision-makers in order to improve the overall efficiency of any optimizations. A distinguishing aspect of KDO is that the meaning of knowledge-driven is two-fold: (1) perform optimization “driven” by the need to generate knowledge in terms of rules, patterns, sensitivity, and meta-models; (2) gather knowledge that can be utilized to “drive” optimization. The KDO framework requires (1) the generation of knowledge from optimization, mostly MOO data, and (2) the use of knowledge to drive optimization. These requirements can be met through two closely related algorithm types, namely, (1) knowledge discovery algorithms and (2) evolutionary algorithms (EAs). Knowledge Discovery involves the use of DM and/or machine learning algorithms that are capable of building knowledge models from optimization data.

Most traditional DM methods are unsuitable for this purpose because they treat all data features alike. Optimization data is unique because it contains both the values of decision variables and objective functions, which must be treated differently: decision variables represent what a decision-maker can adjust, while objective values reflect the outcomes of those adjustments. Decision-makers rely heavily on objective values to make informed decisions. Therefore, an appropriate DM technique should be able to extract knowledge about the variables that significantly influence these objectives. Existing general-purpose DM methods often require adaptation before they can be effectively used to examine data from optimization tasks. EAs are among the most widely used MOO methods in practical applications. Rooted in the principles of natural evolution, they are known for their effectiveness in solving real-world optimization problems, including those with large size, non-linearities, stochasticity, and non-analytic characteristics. However, the most significant advantage of EAs over traditional optimization algorithms lies in their ability to handle multiple optimization objectives. For comparison to EAs, we can

consider the field that has been referred to as design space exploration, a method for specialized DM with applications primarily focused on product design. Design space exploration has been investigated in studies, such as [20] and [21]. Other studies integrating DM, simulation, and optimization can be found in [22], [23], and [24]. However, these studies have predominantly addressed single-objective optimization problems.

B. CONTRIBUTION

To further support human decision-makers, group decision-making, and the future use of virtual manufacturing through VF-KDO, we have performed a literature study to identify the current research related to the use of graphs in this context. Our primary contribution is to identify related research with the potential to support the VF-KDO with the KG concept and gaps in the current research that the VF-KDO concept can fill.

To guide this work, we defined four questions for this review to address;

- 1) Where and how are graph and semantic technologies currently being considered in industrial research?
- 2) Where and how are graph and semantic technologies being used in conjunction with optimizations (Knowledge (graph) Driven Optimization)?
- 3) Where and how are graph and semantic technologies being used to support human decision-making in the industry?
- 4) Are graphs being used in industrial contexts to capture and trace the provenance of decision-making with relation to optimization activities, and if so, how?

In the following sections, we first describe the methodology of our literature review in more detail, describe the execution of the review, and the process of filtering and selecting papers from those provided by a range of search engines. We then present a more detailed analysis of the papers that were selected for the final review and the patterns that are apparent.

II. METHODOLOGY

In conducting this review, it was deemed essential to adhere to a structured and systematic methodology for the identification and selection of relevant literature. In addition to categorizing the literature statistically, we also sought a deeper understanding of each paper’s contents. To achieve this, the final stage of the review involves a more traditional qualitative review of a selected subset of the identified literature. To this end, we have used an existing hybrid systematic-narrative review process as our template [25], which is itself a hybridized version of several other approaches. The template suggests that the following stages are common to different kinds of literature review: (i) the Question(s), (ii) the Source(s), (iii) the Selection process, (iv) the Evaluation process, and (v) the final Synthesis. A hybrid literature review process should carefully consider all of these stages.

A. QUESTIONS

We began this investigation with the general question of the current state of research into the application of graph databases and technology in industry, with a particular emphasis on decision support. The four questions we developed from this broad topic have been previously stated in the introduction.

B. SOURCES

The sources and the collection of potential papers were conducted in two phases. In the first phase, we selected a set of simple keywords that we felt supported the research questions. The keywords were then used to search the IEEE and Science Direct databases. These two capture a good range of publishers and support our particular fields of Industrial Information Systems and Engineering. These broad queries often resulted in impractical numbers of papers (many thousands), and where this happened, we kept the top 100 results. The results of these queries were glanced through to give a feel for the types of publications and the types of keywords that were being used in preparation for the second phase.

After the first collection phase, a query was crafted to more carefully target the specific interests that our questions raise. This query was then applied to a broader group of databases to ensure comprehensive coverage of the topics of interest. Not all the databases support precisely the same query. Where this was the case, we followed the intended query as best we could and recorded the changes on a case-by-case basis.

C. SELECTION AND REMOVAL OF PAPERS

The results of all the searches described were combined into a single pool of papers for examination. The first stage of this process was to remove papers that were not necessary to the review, including duplicate papers (identified by title, author list, and date) and removal of papers that did not seem to be directly relevant based on the title and abstract (for example several papers on deep neural networks refer to semantics, but this is not what we intended in our original keyword search). At this stage, we removed papers not written in English from the final set.

The next stage involved a cursory reading of the remaining papers to decide their relevance and categorize them according to their broad content. We identified three different ways to categorize each paper: (i) based on the industry they focused on (if any), (ii) the approach taken (for example, were they trying to create KGs or were they focused on usage and if so, in what way), and (iii) the topics that the paper addressed (for example, did the paper focus on data integration between devices or did the paper focus on planning in factories). We chose to re-classify the papers rather than rely on the author's keywords to represent how we saw the paper within our field of interest, which was not necessarily captured by the reported keywords. These classifications and notes formed the basis for our later groupings and classifications.

Once this initial removal and classification process was complete, we had a set of 101 papers. To provide a narrower set of papers for a more in-depth qualitative review, we utilized the tags we had previously assigned as content indicators. We approached this as a set-covering problem, aiming to select a subset of papers that provided comprehensive coverage of all topics. To execute the selection process, we used a genetic algorithm¹ implemented using Pymoo [26]. To maintain a minimum breadth of the study, we added a rule to the algorithm that the number of selected papers should be more than 30, though it turned out this was not necessary.

To further strengthen the selection process, we also integrated data from the Norwegian list [27] to highlight which papers were from scientifically respected publications (using 2023 ratings) and data related to how well-cited papers were from Crossref [28]. A weighted objective function encoded all the rules for the optimization process: each of our categories must be presented by at least one paper; use the smallest number of papers; prefer papers with higher numbers of citations; and prefer papers from publications with a higher value in the Norwegian list.

D. EVALUATION AND SYNTHESIS

To perform a broad bibliometric analysis and give an overview of the interest in, for example, the range of industries that researchers are investigating, we made use of the set of 100 papers identified before our focusing selection. The illustration of the category breakdown was then carried out, and these results can be seen in Figures 2, 4, and 3. These statistics were then used to provide insights into our four key research questions.

Beyond the bibliometric analysis, the papers selected via the genetic algorithm were re-examined, and this detailed consideration informed our final synthesis and analysis.

III. IMPLEMENTATION OF THE REVIEW

In order to balance breadth and specificity in answering the research questions posed, two search tasks were carried out to provide the initial pool of papers. In the first search task, several sets of keywords were selected and used to search two well-known databases for engineering and sciences, specifically IEEE and Science Direct. In the second search task, a more complex query was constructed and applied to five databases: Scopus, IEEE, ScienceDirect, Dimensions, and Springer. Each search yielded many results, with the top 100 retained for further analysis. The queries, necessary database-specific modifications, and the number of results obtained from each database for this survey are presented in Tables 1 and 2.

Once the papers had been collated, a standard series of filters was applied, including identifying duplicate papers and filtering papers based on their titles and abstracts. We did not

¹We are aware that exact methods for this problem exist. However, we sought a reasonable guide for where to focus attention rather than a mathematically perfect result.

TABLE 1. The results of the first search task with the keywords shown for each database. A maximum of 100 results were exported for further analysis.

Database	Keywords of Query	Result
IEEE	knowledge graph production industry neo4j production industry semantic production industry	80 3 100
Science Direct	knowledge graph production industry neo4j production industry semantic production industry	100 100 100

TABLE 2. The results of the second search task with the query; Title and Abstract - ("Knowledge Graph*" OR "Neo4j" OR "Semantic" OR "Graph Database*") AND ("Production" OR "Manufacturing" OR "Industry").

Database	Database-specific Modification	Result
Scopus	Limited to appropriate subject areas, articles and chapters in English.	58
IEEE	Limited to conferences, journals and standards.	76
Science Direct	Limited to appropriate subject areas and article types in English	59
Dimensions	Limited to appropriate subject areas and article types	68
Springer	Queried for both Conference and Article separately	133

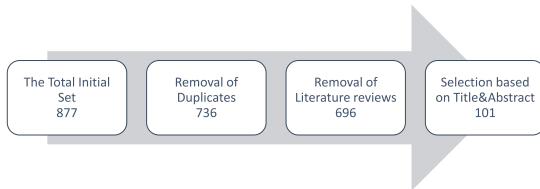


FIGURE 1. A diagram of the impact of our selection process on the number of papers during the literature review.

directly include other literature reviews within our own, so we removed the 40 literature reviews we found and examined them separately. The details of this process are presented in Figure 1, including the number of documents that were separated or removed at each stage.

This process left us with 101 papers, which were still too many. Before applying the genetic algorithm, we first examined the papers to identify the broad patterns of the themes. The full breakdown of the categories can be seen in Figures 2, 3, and 4. Most papers were identified to belong to one grouping for each top-level category (industry, topic, and approach). However, not all papers did fit naturally.

At this stage, the genetic algorithm was employed to achieve a set covering of the categorized literature, as described in our methodology. Once the genetic algorithm was completed, the number of papers from the search results was reduced to 43. A detailed breakdown of the higher-level categories and how the papers are placed in each category can be found in Table 3.

TABLE 3. The breakdown of the categories we have created and the papers we have selected.

Category	References
Approach	Analysis [29]
	Applications [30]–[41]
	Construction [35], [41]–[50]
	Queries [44], [48], [49], [51]–[56]
	Applications [57]–[60]
	Construction [61]
	Ontology [37], [38], [50], [53], [57], [62]–[65]
	Queries [38], [52], [53], [57], [58], [62], [63], [66]–[70]
	[50]
	Food [37]
Industry	Forestry [64]
	[34]
	Pharmacology [54]
	Construction [38], [53], [56], [65], [68]
	Energy [51]
	Mining [41]
	Petrochemical [48], [62]
	IT [29]
	Other [71]
	Automation [36], [39], [59], [66], [67], [70]
Topic	Design [43], [44], [46]
	Machining [52], [57]
	Manufacturing [32], [33], [58], [60]
	Materials [45], [59], [61]
	Other [30], [35]
	Processes [55]
	Smart [31], [40], [49], [63], [69]
	Sustainability [47]
	Air [42]
	Shipping [35]
Simulation	Design [44], [56], [65]
	[64]
	Analytics [63]
	Integration [31], [49], [50], [57], [59], [66], [69], [71]
	Knowledge [42], [48], [51], [55]
	Machine Failure [34], [41], [43], [45], [70]
	Open Data [50]
	Traceability [32]
	Faults Safety Security [29], [36], [39], [55], [62]
	Flexible Manufacturing [60]
Planning	Forecasting [52], [68]
	Life Cycle [33], [58]
	Optimization [30], [40]
	Project Management [38]
	Requirements [46], [61]
	Resource Matching [35], [47], [54]
	[60]
	Digital Twin [37], [53], [67]

CONSIDERING LITERATURE REVIEWS

The initial set of papers retrieved by the database search yielded 40 papers that were identified as literature reviews. Many of these papers were irrelevant to this work or the VF-KDO concept. Several addressed very general subjects, such as KGs, instead of a more specific interest in industrial applications of KGs. Others were narrowly focused but on other business sectors, such as social networking. Many others addressed concerns that were not directly relevant to this paper, such as literature reviews on big data. Based on their titles, we selected three papers that focus on topics similar to those we are interested in, specifically industrial production systems and KGs [72], [73], [74]. Notably, these literature reviews are relatively recent, the oldest being from 2018 and the others from 2021. All three have been published in journals or conference proceedings of reasonable scientific quality or recently launched, resulting in the absence of a rating.

IV. ANALYSIS AND DISCUSSION

A. LITERATURE REVIEWS

The first two literature review papers [73], [74] both look at device interoperability and place a special emphasis on

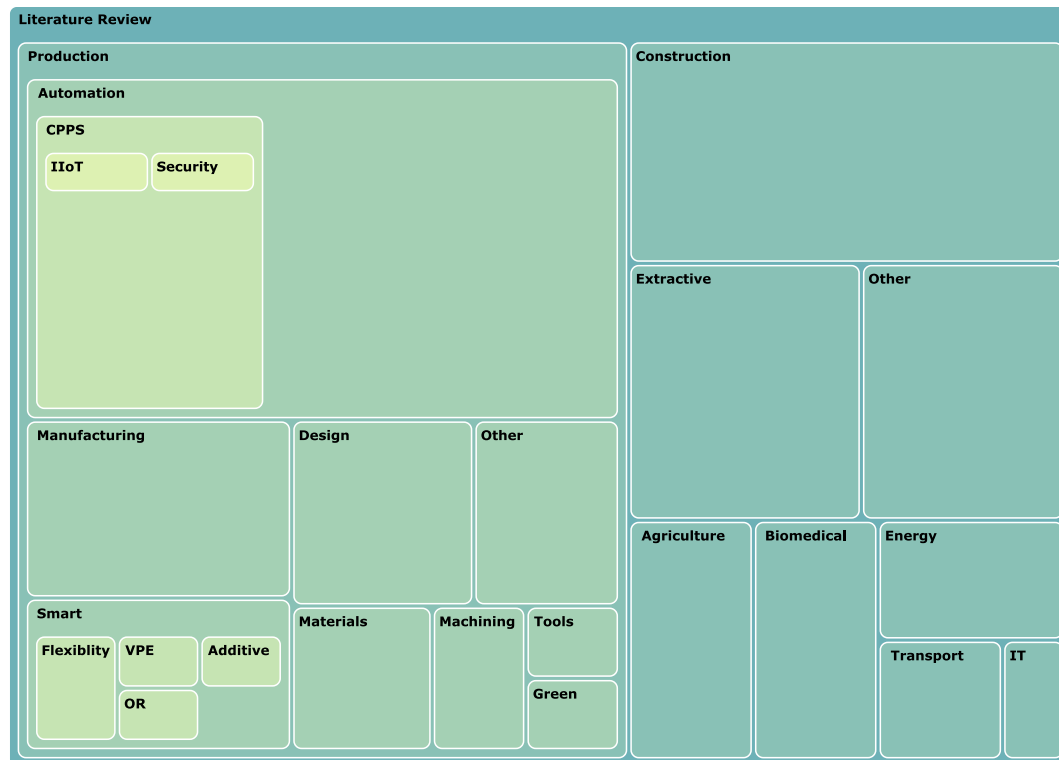


FIGURE 2. Tree-map chart showing the grouping and categorization of the broadest group of papers that we considered, based on industry. The area of each box represents the relative number of papers included in that category and those below it.

integrating heterogeneous data sources. Integrating heterogeneous data is presented as a key concern in realizing the goals of Industry 4.0, modeling devices in a standard way, and providing a standardized way to model and gain access to data. Both papers discuss the Internet of Things as an enabling technology. Of the two, the earlier paper by Nilsson and Sandin [73] is more general and mentions various application domains, including construction, healthcare, and production. The later paper by Beden et al. [74] is focused on interoperability between industrial devices through the use of Asset Administration shells. Both papers emphasize the use of the semantic web technologies of RDF and OWL to provide semantic models of devices and data to enable data integration. This approach allows the devices and data of a set of factory devices to be seen as a KG and, at least in principle, queried as one.

These two papers, separated by three years, discuss several example applications and research projects. Noticeably, in both papers, the examples seem not to be large-scale, consisting of at most a thousand devices and usually far fewer. Nilsson and Sandin [73] even mention that they had not found evidence suggesting that the solutions could scale, though they do not say there is evidence that the solutions cannot be applied at scale. These papers also do not place a great deal of emphasis on the human decision-making process,

though Beden et al. [74] do acknowledge human modeling as an open challenge.

The third literature review we considered [72] is different in several ways; it is a structured literature review similar to the review we have conducted, though it does not focus on Industry 4.0 but on manufacturing generally. It also explores KGs rather than semantic networks and interoperability. The review lists a range of fields where KGs have been applied, including healthcare and tourism. The authors also suggest that, compared to other research areas, such as AI and neural networks, there is a relatively small interest in KGs for industry. They also discuss where the application emphasis seems to be placed in the papers they have reviewed. According to the authors, the largest area of interest has been *search-based applications*, where the emphasis is on analyzing the relationships between data and with a focus on *analytical tasks and knowledge discovery*. The other categories they discuss include cyber-physical systems, data representations of manufacturing and production processes, simulations, and the construction of KGs. Of these, the analysis of data and knowledge discovery and the integration of KGs and simulations are the closest to our interests in VF-KDO. A closer look at the papers referenced in [72] shows that they discuss KGs that store data about simulations and store data from sensors, devices, or processes in

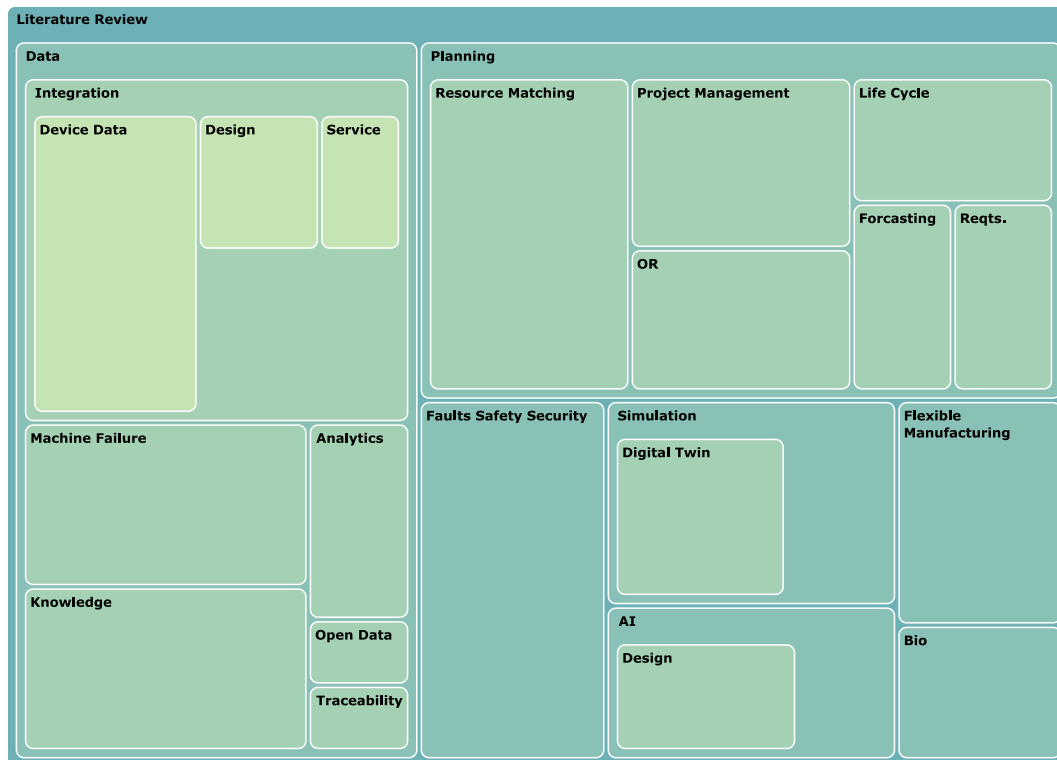


FIGURE 3. Tree-map chart showing the grouping and categorization of the broadest group of papers we considered, based on the topics addressed in them. The area of each box represents the relative number of papers included in that category and those below it.

the factory. These findings are in line with the results that we also present here.

B. BIBLIOGRAPHIC DISCUSSION

We begin with a brief review of when the papers considered in this review were published. Figure 5 shows a breakdown of the papers considered in both the broader review seen in the tree-map breakdowns (Figures 2, 3, and 4) and the 43 papers selected to present coverage in the final discussion. The chart demonstrates that the pattern of publications is relatively similar between the two selections. The majority of the papers were published since 2016, with a small selection published between 2009 and 2010. Of particular interest is that the selection of papers by the GA included two-thirds of these papers, suggesting they address specific industries or topics not addressed by other, more recent papers.

Figure 2 shows the papers' breakdown by the industries upon which they focus. The search terms we used in the initial search focused on production and manufacturing, as they fit the interests of the VF-KDO concept. Hence, it is not surprising that a majority of the papers were found to have a focus on production. More interestingly, many other sectors are present, particularly the construction, mining, and petrochemical (extractive) industries. Several papers related to the construction industry were selected for the final analysis, and they show an interesting focus on using KGs to manage the documents related to buildings,

including Building Information Models (BIM) [38], [56], [65], [68]. These will be an interesting source of inspiration for our later work. The agricultural and bio-medical sectors are also present in the breakdown, though to a considerably more limited degree.

The emphasis within the production subcategory is on the field of automation. This subtopic includes work on the Industrial Internet-of-Things, data gathering, and integrating information from multiple machines. However, generally, this category includes papers that address some sort of factory automation. Also included in the production sector were papers discussing manufacturing processes, such as those found in the semiconductor industry, though these did not have the same focus on the issues of data gathering. We also found some papers discussing simulation, optimization (smart industrial concerns), design, and papers focusing on specific materials' processes.

Figure 3 shows a breakdown of the papers based on the core topic (often a technological topic) that those papers have focused on. These classifications combine both the problem that is being addressed, such as understanding machine failures [34], [41], [43], [45], [70], but also the underlying technology that is being researched, such as optimization [30], [40].

Within the production subcategory, we see a broad range of interests. The largest grouping focuses on data, how devices

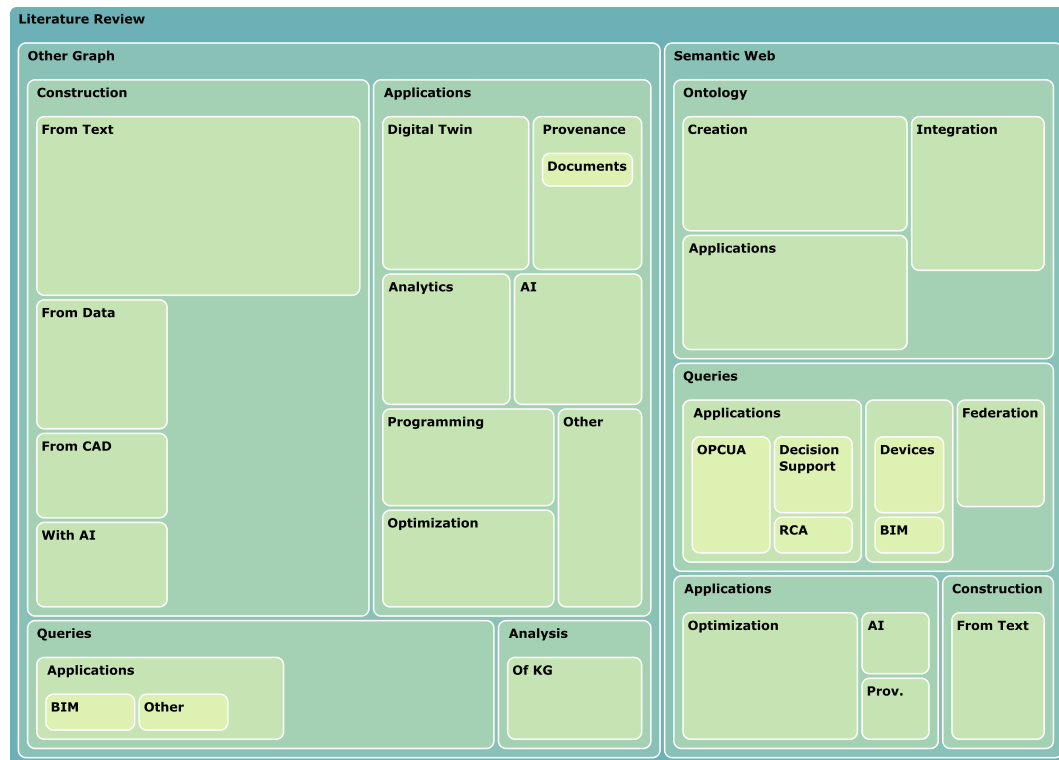


FIGURE 4. A tree-map chart showing the grouping and categorization of the broadest group of papers we considered, based on the approach adopted in them. Note that these terms need not overlap with terms in other tree-maps. For example, ‘construction’ here refers to the paper that focused on how to construct KGs. The area of each box represents the relative number of papers included in that category and those below it.

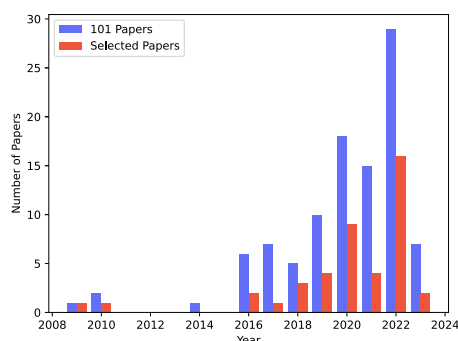


FIGURE 5. The distribution of papers and their published years, including the complete set of 101 papers and the subset of papers selected by the set covering genetic algorithm. This indicates a broad alignment between the scale of publishing in the broader set and those selected to highlight the findings.

present their data, and how to utilize graphs and semantic technology to integrate data within the factory. The second largest grouping is papers that discuss issues related to planning. These papers cover a range of concepts, including optimization and the related issues of resource matching. However, papers interested in project management and product life cycle analysis are also included.

The remainder of the papers discuss a range of topics, including AI, flexible manufacturing, simulations, and a larger group interested in fault, safety, and security analysis.

This is more than machine failure, with security often referring to IT security analysis, while fault and safety refer to the analysis of machines, production processes, or construction projects with an eye toward avoiding faults and improving safety.

Figure 4 shows a breakdown of the papers based on the approach taken to utilizing graph databases. This can overlap with the *industry* and *topic* breakdowns. For instance, where the paper’s emphasis is almost entirely towards a specific application, it can overlap with the *industrial* classification. One example is the emphasis on BIM within the *Querying* classification, which can be seen in the sampled paper [38]. The overlap with *topic* is similar and appears where the approach being taken places so much emphasis on a topic (such as AI) that the application becomes the core of their approach.

Overall, the papers are divided into two groups: those that emphasize the Semantic Web stack of technologies [75], and the slightly larger group that use other forms of graph technology or are unclear about the technology they use. Within both groups, some subcategories emphasize the building of KGs. This is the smallest group in the semantic web category, although papers discussing ontology creation may also be included. Outside of the semantic web category, it is the largest subcategory. This interest in the automatic creation of KGs from other forms of data is also seen in recent

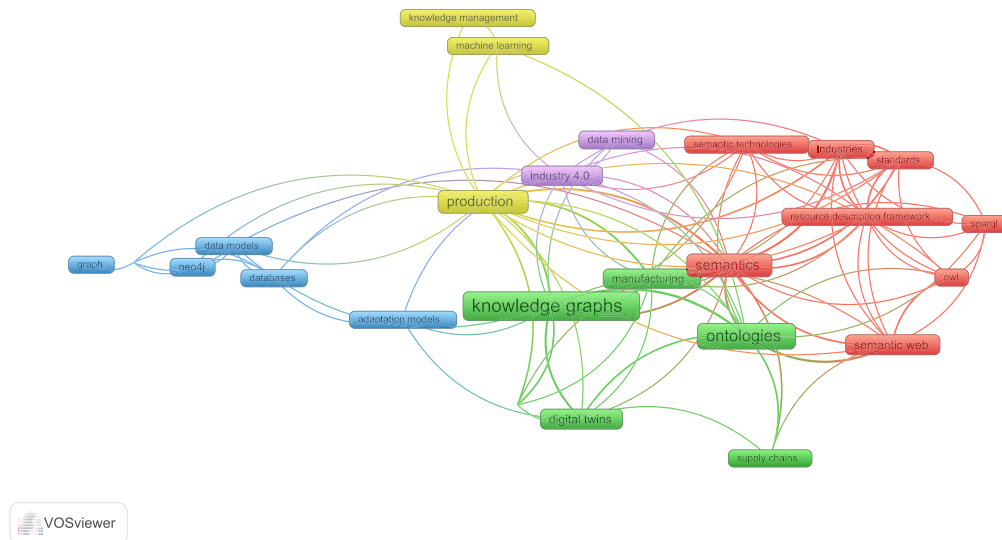


FIGURE 6. A chart showing the co-occurrence of keywords in the papers selected, and the number of papers utilizing each keyword. Generated by VOS Viewer [77].

developments within Neo4J, where they are integrating Large Language Models (LLMs) for exactly this purpose [76]. There is also a general branch of research into how querying is best implemented for specific applications and in terms of the algorithmic execution of queries on graph databases.

Of particular interest to the VF-KDO concept is a group of papers that address the provenance of information stored and examined using graphs [31], [32], [38]. While their specific interests vary, we believe that these will provide a strong basis for our later research.

C. KEYWORDS OF OUR HIGHLIGHTED PAPERS

While we have not relied on the keywords associated with the papers for our direct analysis, we have analyzed and clustered the 43 highlighted papers using VOSViewer [77]. The keyword co-occurrence analysis is presented in figure 6. The presence of a keyword in this graph indicates it appears in the keyword list of at least two papers. We must note that we manually modified the keywords to reduce confusion, for example, by combining “knowledge graph” and “knowledge graphs” into one group.

Three significant groupings appear: graph/databases (blue), KGs (green), and semantic web (red). A fourth category related to AI, including data mining and machine learning, is present above the others (yellow and purple). We can also see that the KG and semantic web clusters are more tightly interconnected, while the graph/databases cluster is more isolated from the rest of the diagram.

While this chart seems to confirm our categories generally, many details we were interested in are not visible. This includes the specific types of industries focused on in different papers and the types of applications for which the technologies are used (although “supply chains” is quite prominent, it seems to be the only one).

D. SUSTAINABILITY

While conducting our primary analysis, the impact on sustainability was not identified as a core interest in the papers. Despite this, we believe it warrants additional focus, and therefore, a further examination was conducted. What we noted was that, while many papers considered issues of industrial efficiency and, in that sense, could be argued to be interested in various forms of sustainability, very few papers directly addressed it.

The papers that addressed sustainability most directly typically did so through the application they were addressing, such as forestry and fire prevention [64] and urban planning [53], [65]. Similarly, [58] considers environmental impact through the lens of product end-of-life, and [33] explicitly addresses using KGs to enable sustainability in life-cycle analysis. Other papers have only touched on sustainability, however, not in direct connection to KGs. These included the application areas of agriculture [50], [54] and business sustainability [30].

In general, the papers we have examined remained within their field of specialization and did not address issues of sustainability in their various industries, nor the question of how directly KGs themselves and the computational infrastructure they require are sustainable.

E. HOW DO THE PAPERS SUPPORT THE VF-KDO CONCEPT

The VF-KDO concept is interested in improving production systems by integrating knowledge generated through simulation-based optimization and developing software tools to support this. Within the selected papers, several are aligned with our interests, each offering a distinct perspective.

The first perspective we will consider is an emphasis on simulation. This is divided into two parts: those that

discuss simulation in general and those that place a special emphasis on digital twins. Within our narrowed survey, three papers discuss digital twins, and only one other focuses on simulations (though an argument can be made for those papers also covering optimization, of course). Of those that discuss digital twins, two present graphs as a way of storing the structure of the twin itself and the relationships between components. Interestingly, one is on the massive scale of smart cities [53] and, as such, is of less interest to VF-KDO due to the specialized nature of the geospatial relationships that they are modeling. The second paper to discuss the storage of digital twins describes them as *cyber-twins* and looks at how to store a standard machine model in a graph database and then generate the digital twins from them, hence allowing knowledge reuse [37]. The final paper on digital twins discusses using the graph to model the surrounding context of the digital twin, such as the local atmospheric temperature [67]. This use is less about knowledge sharing and reuse and more about using graphs to manage and share real-time data about devices. Within this group of papers, we can see an interest in using graph structures to store models of industrial systems, but at present, they show less interest in linking graph structures with optimization or the evolution of simulation models as they are adapted over a system's development or lifetime.

The second perspective is the relationship between simulations and ontology, which is directly addressed in [60]. This paper considers the creation of a tool that, through an ontology (or KG) and graph models of previous simulations, can derive new specialized simulations more quickly than a researcher starting from scratch. This sharing and reuse of simulation-related knowledge is important to future work within the VF-KDO concept. Notably, this paper indicates that work on this topic is nearly as old as the concept of the semantic web itself, such as the work in [78], discussing the potential of using KGs to manage simulations.

The next perspective is optimization, which is linked to the VF-KDO concept of studying the evolution of simulation models with respect to optimization and how one impacts the other. Two papers in our survey have directly discussed optimization. One utilized Neo4J as a knowledge store for a more flexible and detailed representation of the Job-Shop Problem and retrieved the specifics of an instance of the problem for processing using Ant Colony Optimization [40]. A similar approach can be seen for resource allocation in logistics [30], where details of the environment's layout are stored in a graph. In this example, the authors make use of the shortest-path graph algorithm to enable the matching algorithm. However, it is unclear if they are utilizing it within a graph database or downloading the data for external processing.

Resource matching can also be considered a form of optimization. Several papers have shown interest in this application [35], [47], [54], and in each case, the focus is on KGs to enable the finding of interconnected information quickly.

For example, [54] proposes a KG of plants, their chemical components, and processes that can transform these chemicals into other chemicals. The industrial interest here is in quickly working out the potential use of a plant as a source of a target substance for a process. It becomes possible to identify different chemical pathways to target chemicals for a given plant, streamlining decision-making regarding materials.

The final perspective is that of provenance, which means that we want to be able to trace the decisions that led to a given outcome. We consider this to be of great value in supporting group decision-making while working with KDO. Several papers in this literature review have raised provenance, but with different objectives. Often, the interest is in tracing the production of a specific product and the processes and machines used. Graphs provide a powerful way to integrate and model the heterogeneous data found in this problem, as seen in [32]. As discussed in the industry section, many authors have examined managing documents describing buildings. Of particular interest to us is [38], where they examine the management of documents in a project related to buildings and include the concept of the provenance of individual features of the final model and how an element came to be included. Finally, and closest to our work is [31], where the concept of using the KG to manage designs from Computer Aided Engineering in a data lake is considered. Their approach includes multiple stages of the evolution of a virtual design, including 3D modeling for materials and simulation with the results of that simulation, and being able to backtrack multiple sets of simulation results to the same original geometry.

We also note that much work is being done on other ways to use, process, and build KGs. For instance, several papers have discussed building KGs through the processing of large numbers of text documents, which can then provide a resource for understanding the business or a way to examine similarities between industrial processes. For example, [42] provides an example of using AI to examine the documents of industrial processes in an organization and then provides a single unified KG of all the links between them. Similarly, researchers in the field of construction technology have done work processing digital models of buildings and capturing the interlinking of these documents [38], including on very wide scales and using this to support planning and decision making [65]. Closer to the field of production is work on storing graph models of digital twin models, and being able to show how a single device, or device type is shared in different twins [37]. The relevance to KDO is that of enabling a broader connection, through the KG, to process and industry knowledge stored within business documents.

V. CONCLUSION

In this paper, we have conducted a comprehensive review of the existing literature on the subject of KGs, with a particular focus on their industrial applications. This is partly due to the growing interest in KGs in general and for industrial research

in particular. However, our research has been driven by a more specific interest in the VF-KDO concept, which deals with understanding how knowledge gained by optimization of industrial systems through virtual factory simulations can enhance the decision-making process. An important aspect of this is the development, within a project team, of a shared understanding of how a system is capable of behaving and what properties of a system are of most significant importance. This knowledge, or understanding, may not be present in the decision-making team itself, especially when starting a VF-KDO oriented project.

This paper has sought to support these research directions through an in-depth investigation of graph databases and semantic web technology and their current use within industrial research, based on a set of four research questions. To answer these questions, we have examined a wide range of papers found through searches of a number of key databases and reduced the pool through a series of stages. These stages included the removal of duplicates and the removal of papers that did not correspond to the subject of the use of graph databases in engineering applications. The papers that remained were examined in considerable depth and classified into three major categories: industry, topic, and approach. Using this methodology, we sought to address our original research questions, and the main conclusions of this review are as follows;

- 1) Graphs and semantic networks are being researched in a wide range of industries, including production systems (with an emphasis on industry 4.0, cyber-physical systems, and smart production), the architecture and construction industries, agriculture, and the bio-medical field. It is also interesting to note the range of purposes that are being investigated. For example, the construction industry seems to place a great emphasis on document integration and the relationships between different building models. Other research has been on the use of KGs to provide powerful question/answer and support systems for human workers. In a similar vein, KGs to provide a detailed understanding of options for a given problem and context are being investigated, such as for the processing of waste products back into more useful resources [47].
- 2) In our examination of the literature, we have found examples of graph-based technologies used for optimization ([30], [40]). However, in these cases, the graph database is not used as a KG to capture knowledge about the problem. Either the graph is used as a data storage method for the model to be optimized, or to use queries on such a model to simplify the construction of the optimization algorithm. Related to optimization is simulation, and few researchers have considered the connection between KGs and simulations, including for deriving new simulations from old [60], [78].

- 3) In one sense, graph technology is exclusively used to support decision-making. In another sense, the focus of research is usually on other interests, such as enabling machine communication and, hence, the integration of devices in the factory or supporting AI. Project Management and Forecasting were the two classifications (in topic) where KGs were used to support projects or group work. These accounted for 10% of the 101 papers we considered in the broadest group.

It should also be noted that we classified some papers as being interested in Knowledge management. These KGs certainly support human decision making, but were more interested in building archival knowledge of specific topics.

- 4) A few papers discussed provenance within graphs from different perspectives, including architecture projects and the evolution of designs, the tracing of production, and the managing of computer-aided engineering, including simulation modeling [31]. This last seems most important to us as it shows strong similarities to our own proposal, though it does not include the optimization phase and related uses of knowledge that VF-KDO is interested in.

We conclude that while many of the techniques that we are investigating, such as simulation, optimization, knowledge discovery, and provenance of decisions, have previously been examined in the context of KGs, to our knowledge, no single study includes them all. With the need to continue to improve techniques in engineering and production, it seems clear that developing virtual tools that explicitly integrate these fields to better support decision-making and group decision-making has potential benefits.

We propose that our VF-KDO concept will use the integration of these fields as the foundation for strengthening our industrial design and planning capabilities. At the core is the KG encapsulating relevant information, relationships, and contextual insights gained from KDO and/or MCDA (Multi-Criteria Decision Analysis) methods. The graph will provide a dynamic repository accommodating diverse, heterogeneous data sources and facilitate a holistic representation of the manufacturing system design domain. KDO techniques will then be employed to leverage the structured knowledge within the graph so as to guide further optimization processes. This approach transcends traditional optimization by both seeking mathematically optimal solutions and generating valuable knowledge throughout the dynamic group discussion and decision-making cycles to encourage organizational learning and systematic knowledge management.

Currently, GDM research has not focused on KG technology, and we believe that the dynamic interplay between KDO and KG can bring several advantages to the GDM process. Firstly, it fosters a shared understanding among group members by providing a visual and interconnected representation of process and knowledge through the KG.

Secondly, the integrated framework supports efficient solution exploration, enabling analysts and members of a decision-making hierarchy to interactively navigate and search the graph to retrieve previous experience and uncover useful insights crucial for decision support. Furthermore, the semantic clarity enforced by KG ensures consistent terminology and reduces the risk of misunderstandings during group discussions. In terms of VF, manufacturing entities such as machines, resources, and storage, along with their attributes and relationships, are semantically represented in the KG with a VF ontology. Specifically for the KDO aspect, new or revised optimization objectives, decision variables, and constraints are identified based on insights derived from previous KDO studies stored in the KG. This process encourages experimentation and analysis of optimizations while supporting alignment with the specific characteristics and requirements of designing or reconfiguring manufacturing systems targeted by VF-KDO.

Finally, and equally important, the use of KDO-KG enhances the transparency of GDM processes. Group members will be able to trace the provenance of decisions back to the underlying knowledge, ensuring accountability and fostering a deeper understanding. For decision-makers to accept the extracted knowledge, they must trust its sources and the extraction processes, which can be built through verification and reuse over time. While this trust-building process may take time, the KG approach can significantly aid by improving transparency and traceability, ultimately fostering trust in the long term.

This literature review has provided a solid foundational understanding of the current research trends and the potential uses of KGs for modeling relationships between different versions of simulations and the connections between optimization results and specific parts of simulations. Thus, it enables the tracing and sharing of the developing understanding of the problem being solved. This is the scope to which we will now address ourselves in our future research.

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